MTH211: Theory of Statistics
2025-26 (II)
Lecture 3: Some Standard Distributions

A. Discrete Distributions

1. Degenerate or 1-pt Distribution $P(X=c)=1$, pmf $p_X(x)=\begin{cases}1 & \text{if } x=c\\ 0 & \text{otherwise}\end{cases}$

so that cdf $F_X(x) = \begin{cases}0 & \text{if } x<c\\ 1 & x\geq c\end{cases}$ mgf $M_X(t)=E(e^{tX})=e^{tc}$, with $E(X)=c$, $Var(X)=0$, etc.

2. Bernoulli Distribution (a spl. case of a 2-point distribution)

$X=\begin{cases}1 & \text{with prob. } p \quad \text{(usually called a 'success' prob)}\\ 0 & \text{wp } 1-p\end{cases}$

with $M_X(t)=pe^t+(1-p)$, $E(X)=p$, $Var(X)=p(1-p)$, etc. [Note: pmf $P(X=x)=\begin{cases}p^x(1-p)^{1-x}, & \text{if } x=0,1\\ 0 & \text{ow}\end{cases}$]

3. Binomial Distribution
Repeat n independent Bernoulli trials with success probability p.
Define the random variable X = no. of successes in n trials, then
$X\sim$ Binomial (n,p), with pmf

$$P(X=x)=p_X(x)=\begin{cases}\binom{n}{x}p^x(1-p)^{n-x}, & x=0,1,\dots,n\\ 0 & \text{ow}\end{cases}$$

mgf $M_X(t)=\{pe^t+(1-p)\}^n$, directly from that of (2). Why?
with $E(X)=\frac{d}{dt}M_X(t)\big|_{t=0}=np$, $E(X^2)=\frac{d^2}{dt^2}M_X(t)\big|_{t=0}=\dots$, $Var(X)=np(1-p)$, etc.

4. Negative Binomial Distribution

(a) Repeat independent Bernoulli trials with success prob. p till there are r successes.
Define r.v. X = no. of failures preceding the r^{th} success.
Then $P(X=x)=\binom{x+r-1}{x}(1-p)^x p^r$, $x=0,1,\dots$

and $X\sim$ Neg. Bin (r,p)
(Note: the name comes from the generalized binomial series, viz, $\binom{x+r-1}{x}=(-1)^x\binom{-r}{x}$, the negative binomial coefficients from expansion of $\{1-(1-p)\}^{-r}$).
Here $E(X)=\frac{r(1-p)}{p}$, $V(X)=\frac{r(1-p)}{p^2}$, etc.

(b) An alternative representation:
Define r.v. Y = no. of trials required to get the r^{th} success, then
$P(Y=y)=\binom{y-1}{r-1}(1-p)^{y-r}p^r$, $y=r,r+1,\dots$
Here $Y=X+r$, notationwise, as before $Y\sim$ Neg. Bin (r,p)

(c) Special case $r=1$ in case (a). (or, in (b))
$p_X(x)=(1-p)^x p$, $x=0,1,\dots$
is the Geometric Distribution with 'lack of memory' property, i.e.,
$P(X\geq m+n\mid X\geq m)$ or $P(X=m+n\mid X\geq m)$ does not depend m.

5. Hypergeometric Distribution

Samples are without replacement from 2 types of objects, e.g.,
A sample of size n to be drawn (WOR) from an urn containing M white and N−M black balls.
Define r.v. X = no. of white balls drawn in the sample.

$$P(X=x) = \frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}}, \quad \max(0, n-\overline{N-M}) \le x \le \min(n, M).$$

What would the prob. if balls are drawn with replacement (WR)?
Here, we use the notation $X \sim \text{Hypergeo.}(N, M, n)$

6. Poisson distribution

$X \sim P(\lambda)$, $\lambda > 0$ with pmf

$$p_X(x) = e^{-\lambda}\frac{\lambda^x}{x!}, \quad x = 0, 1, \cdots,$$

Poisson approxn to Binomial distribution:
as $n \to \infty$, $p \to 0$ $\ni$ $np = \lambda$ finite,
the binomial pmf $p_X(x) = \binom{n}{x} p^x (1-p)^{n-x} \longrightarrow e^{-\lambda}\frac{\lambda^x}{x!}$ (Verify this!).

7. Discrete Uniform distribution

For n possible outcomes, $P(X=x_i) = \frac{1}{n}$ $\forall$ $i = 1(1)n$
spl case, $P(X=x) = \begin{cases} \frac{1}{n}, & x = 1, 2, \cdots, n \\ 0 & \text{ow} \end{cases}$

B. Continuous Distributions

1. Uniform distribution

$X \sim U(a, b)$ or, $U[a, b]$

Then pdf $f_X(x) = \begin{cases} \frac{1}{b-a} & \text{if } a < x < b, \text{ or } a \le x \le b \\ 0 & \text{ow} \end{cases}$

cdf $F_X(x) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{b-a} & a \le x < b \\ 1 & x \ge b \end{cases}$

mgf, $M_X(t) = \begin{cases} \frac{e^{tb} - e^{ta}}{t(b-a)} & \text{if } t \ne 0 \\ 1 & t = 0 \end{cases}$

etc..